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Mid-Term Examination, Business 33801:

(Monday, 13 December 2021)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 10 questions with a total of 90 points possible. You have 90 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination.

Sign your name here.

True-False-Uncertain (28 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 [6 points] True, False, Uncertain? [Explain your answer.] A luxury good is a normal good with an income elasticity of demand that exceeds 1. If incomes double and simultaneously every price in the economy also doubles, then *the demand for a luxury good will more than double*.

Question 2 [6 points] True, False, Uncertain? [Explain your answer.] Consider a competitive market for olives in which there are only 100 orchards available for growing olives. Each of these orchards is only one of two possible types. 60 of the orchards are identical and each orchard owner has a high individual supply elasticity of $\varepsilon^s = 5$, while the other 40 orchards have lower individual supply elasticities of $\varepsilon^s = 2.5$. The high-elasticity orchards are currently providing 20% of the market's aggregate olive supply while the low-elasticity growers produce the remaining 80% of olive supply. The elasticity of demand for olives is -2. *The market elasticity of supply is 4.*

Question 3 [10 points] True, False, Uncertain? [Explain your answer.] The supply of barley shifts outward due to improved weather conditions, which in turn causes the total market expenditures on barley to fall.

(a). **[5 points]** *The demand for barley is inelastic.*

(b). **[5 points]** *Consumer surplus increases.*

Question 4 [6 points] True, False, Uncertain? [Explain your answer.] Ted is risk averse and has preferences that are represented by expected utility. Ted is offered a choice between receiving a certain payment of 80 or a gamble that pays 400 with probability $\phi_1 = 0.75$ and generates a loss of 800 with probability $\phi_2 = 0.25$. Ted strictly prefers the gamble over the sure thing. *Ted's risk premium of the gamble is less than 20.*

Short problems (62 points total)

Question 5 [10 points] The market demand function for coffee is

$$Q_c^d = 10 - 2p_c + p_t,$$

where p_c is the price of coffee and p_t is the price of tea. The market supply of coffee is

$$Q_c^s = 4p_c - 1.$$

The equilibrium price of tea is $p_t^* = 1$.

- (a). [5 points] What is equilibrium price and quantity of coffee?
- (b). [5 points] What is the cross-price elasticity of coffee demand with respect to the price of tea in equilibrium?

Question 6 [8 points] Rebecca is a risk-neutral investor. She is currently the only bond holder in a profitable company, but the company will possibly fail to earn enough profit in the future to pay off its debt to Rebecca. Because she is the only creditor of the company, she gets every dollar the company earns in profits until the value of the firm's debt is paid off. Once the company pays off its debt to Rebecca, she earns no additional returns from the company's profits.

Holding expected profit constant, does Rebecca prefer that the firm choose riskier projects or safer projects? Explain using a concept from class. [Hint: Graph Rebecca's payoff as a function of the company's profits.]

Question 7 [10 points] Sam's company makes "Sam's jollof rice dinner" which is a frozen, pre-prepared meal of a well known and liked Nigerian dish. He has found he can sell as many jollof dinners as he likes at a price of $p = 8$, but because Sam competes against other Nigerian frozen-food companies, he loses all of his sales if he prices at a higher price. Sam's opportunity cost of producing q dinners is given by

$$C^O(q) = 2q + \frac{1}{2}q^2 + 17.$$

His associated unit cost functions are

$$AC^O(q) = 2 + \frac{q}{2} + \frac{17}{q},$$

$$MC^O(q) = 2 + q.$$

Should Sam continue to produce jollof frozen dinners? If so, what is the optimal output? If not, what would his loss have been had he operated?

Question 8 [14 points] Higgins consumes only ale and biscuits. The price of both ale and biscuits is $p_a = p_b = \text{\pounds}1$. (Ale is measured in pints and biscuits are measured in tins.) His monthly available income for purchasing these two goods is $I = \text{\pounds}100$. Consuming ale and biscuits also takes Higgins's time. Drinking a pint of ale consumes 1 hour of his free time; eating a tin of biscuits only consumes 15 minutes of his time. His total available time for consuming ale and biscuits each month is $T = 50$ hours.

(a). **[8 points]** Draw a graph illustrating Higgins's two constraints and Higgins's feasible region of ale and biscuit consumption. Label all important parts of your graph.

(b). **[6 points]** Higgins's has constant marginal utility of consuming both goods. His marginal utility of a pint of ale is constant at $MU_a = 1$, while his marginal utility of a tin of biscuits is constant at $MU_b = 2$. How many ales and biscuits should Higgins consume each month? Explain.

Question 9 [10 points] Consider the market for durian fruit in Singapore. Assume that there are two types of consumers, A and B , that enjoy eating durian. Each week, type- A consumers would be willing to pay up to \$10 for their first durian, \$8 for their second, and \$2 for their third; they do not want more than 3 durian a week. Type- B consumers would be willing to pay up to \$12 for their first durian, \$6 for their second, but they do not want more than 2 durian a week. Suppose that there are exactly 1000 consumers of each type living in Singapore and no one else ever buys durian. The supply of durian in Singapore is infinitely elastic at a price of $p = \$5/\text{durian}$.

(a). **[5 points]** What is the equilibrium price and quantity of durian consumed in Singapore per week? [Hint: Draw the durian supply curve.]

(b). **[5 points]** In equilibrium, what is each type- A 's consumer surplus from the durian market? (No need to compute it for type B consumers.)

Question 10 [10 points] Sonny Camera Company sells two different cameras. The high-end model, the A1, is priced at \$6000; the opportunity cost of all the parts for the A1 except the specialized CPU chip is \$4000. The low-end model, the B2, is priced at \$2500; the opportunity cost of all the parts for the B2 except the specialized chip is \$1000. Both cameras use the same CPU chip, which due to supply chain disruptions can only be obtained by Sonny in limited quantities at a price of \$750. Unfortunately, the demand for each model camera is greater than the number of CPU chips Sonny can obtain, and consumers who don't succeed in buying a Sonny camera will buy a competitor's product instead (i.e., the sale will be lost forever). Taking the prices as given, what is Sonny's opportunity cost of producing the B2 camera during the CPU chip shortage?

Answers to mid-term examination

1 False. If incomes and prices double, the budget line is unchanged and hence demands are unchanged.

2 False. A one percent increase in price leads each low-elasticity orchard to increase output by 2.5 percent and each high-elasticity orchard to increase output by 5 percent. The percentage change in the market's total output is therefore the weighted average (using percentages of outputs) of these responses. Hence, $\varepsilon^s = 0.8(2.5) + 0.2(5) = 3$. The number of orchards in each segment is not relevant once we are told the fraction of total output each segment produces. The elasticity of demand is also not relevant.

3 (a). True. An outward shift in supply causes the equilibrium quantity to increase. Given that the change in equilibrium price and quantity is driven by a shift in supply, both equilibrium points lie on the same demand curve. For total expenditures to also fall requires that $MR < 0$, which is equivalent to being in the inelastic region.

(b). True. Because the price has decreased, the total consumer surplus in the market must increase. That is, there must be more area between the demand curve (which has not shifted) and the market price (which has decreased).

4 (a). True. The expected value of the gamble is

$$E[\tilde{x}] = \frac{3}{4}(400) + \frac{1}{4}(-800) = 100.$$

Because Ted strictly prefers the gamble to a certain payment of 80, his certainty equivalent for the gamble must exceed 80. Hence, the cost of the risk (i.e., the risk premium) which is $\rho = E[\tilde{x}] - CE = 100 - CE$ must be less than 20.

5 (a). Setting $Q^d = Q^s$, we have

$$10 - 2p_c + p_t = 4p_c - 1.$$

Given that $p_t^* = 1$, this simplifies to

$$11 - 2p_c = 4p_c - 1.$$

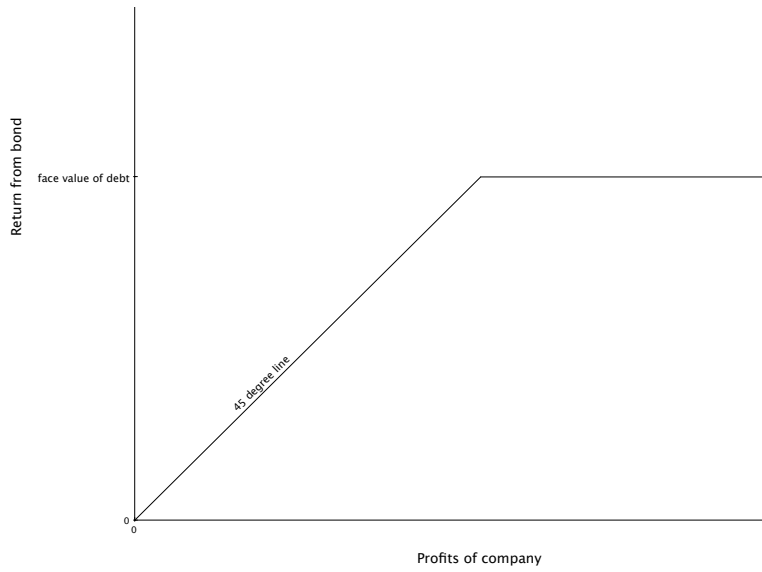
Solving for p_c^* , we obtain $p_c^* = 2$. At the given price, coffee demand is

$$Q_c^* = 10 - 2(2) + 1 = 7.$$

(b). The formula is

$$\varepsilon_{Q_c, p_t} = \frac{p_t^*}{Q_c^*} \frac{\Delta Q_c^d}{\Delta p_t} = \frac{1}{7}(1) = \frac{1}{7}.$$

6 The bond that Rebecca owns results in her obtaining all of the downward movements in profits, but only a portion of the upward movements since she can never make more than the value of her bond. Because the upside is capped and Rebecca is otherwise risk neutral, the bond makes her behave as though she were risk averse. This can be seen graphically; the return from the bond is a concave function of the underlying profitability of the firm.



7 The marginal output rule requires $MC(q^*) = MR(q^*)$. In our setting, we have $MC(q) = 2 + q$ and $MR = 8$. Combining these expressions, we have

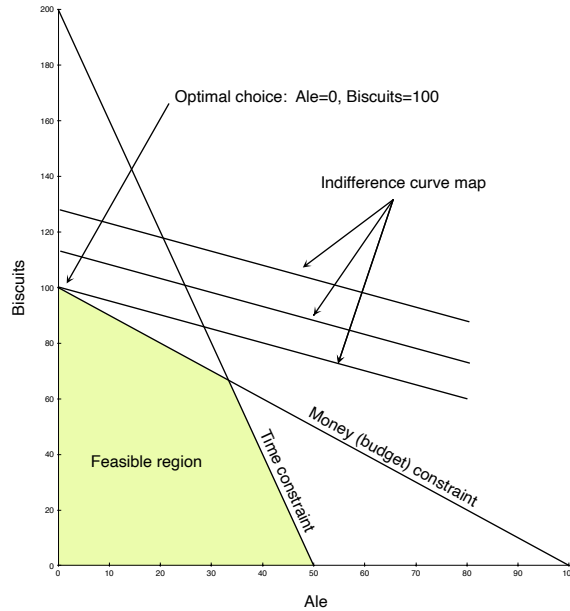
$$2 + q = 8$$

or $q^* = 6$. At this output, economic profit would be

$$\pi = 8 \cdot 6 - (2 \cdot 6 + \frac{1}{2} \cdot 6^2 + 17) = 1 > 0.$$

Hence, profits are positive and the firm should produce $q^* = 6$ units.

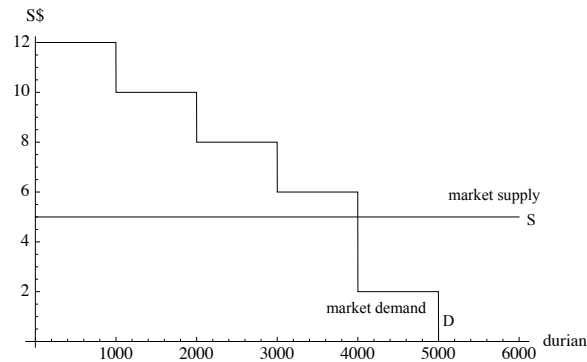
8 (a). Here is the graph with ale on the horizontal axis and biscuits on the vertical axis.



(b). Higgins should spend all of his money on biscuits: $(a^*, b^*) = (0, 100)$. This satisfies both the budget and time constraints. Given his $MRS_{a,b} = \frac{MU_a}{MU_b} = \frac{1}{2}$, and given that this is less than the slope of the budget line (slope $= -\frac{1}{2}$) or the time constraint line (slope $= -60/15 = -4$), he should only be consuming biscuits.

9 (a). Because supply is perfectly elastic (i.e., horizontal) at $p = 5$, we know that the equilibrium price will be $p^* = 5$. At this price, type A consumers will buy two durian each and type B consumers will buy 2 durian each. Given there are 1000 of each type, we have an equilibrium quantity of $Q^* = 4000$.

You did *not* have to draw any graphs for full marks on this question. But if you did choose to draw the market supply and demand curves in order to compute the equilibrium, they should look like this:



(b). Type A consumers would have paid \$10 for their first durian, and \$8 for their second; they only had to pay \$5 for each, however. Thus the first unit produces 5 in surplus and the second unit produces 3 in surplus. In total, type A consumers obtain a weekly surplus of \$8.

10 The use of the CPU chip in a B2 camera means that one less A1 camera can be sold during the chip shortage. The foregone A1 profit is $6000 - 4000 - 750 = 1250$. Hence, the opportunity cost of producing the B2 is the direct cost of \$1000 of the parts plus the cost of the chip, \$750, plus the foregone A1 profit of \$1250. Hence, the opportunity cost is \$3000.